

MUHAMMAD TAIMOOR TARIQ

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Education

University of Illinois at Urbana-Champaign

August. 2022 – May 2027

PhD. in Computer Science

IL, USA

- Advised by: Radhika Mittal

Lahore University of Management Sciences (LUMS)

Sept. 2018 – May 2022

Bachelor of Science in Computer Science

Lahore, Pakistan

- Placed on the Dean's Honors List for academic excellence for years 2020-2022

Experience

Google

May 2025 – August 2025

PhD Software Engineering Intern – Network Hotspots Team

Sunnyvale, CA

- Designed and implemented a C++ simulation tool for novel networking services, enabling estimation of achievable bandwidth under diverse workloads.
- Integrated the simulator with large-scale cluster management frameworks, leveraging production checkpoints and historical usage data for accurate modeling.
- Developed a max-min fairness optimization model to ensure fair and efficient bandwidth allocation across competing services.

Slice-Aware Resource Management For 5G Networks

August-2022 - Present

Research Assistant - Advisor: Dr. Radhika Mittal

- Worked on projects related to Slice-aware Interference Management and Load Balancing for Multi Cell Deployments.
- **Under submission at IEEE SEC 2025**

Fault Tolerant Cellular Control Plane through Replication At The Edge

June 2021 – May. 2022

Research Assistant - Advisor: Dr. Zafar Ayyub Qazi

- Goal was to design a fault tolerant control plane by deploying replicated control plane nodes at the edge
- Implemented and evaluated the system against prior works (DPCM, ECHO, SkyCore) under varying network conditions
- Results showed an improvement of up to 3.8x in Procedure Completion Times under failure conditions

Efficient, Consistent Edge-based Cellular Control Plane

June 2020 – April 2021

Research Assistant - Advisor: Dr. Zafar Ayyub Qazi

- Goal was to minimize latency in cases of user mobility by implementing a geo-aware, proactive state replication system
- Implemented a consistent-hashing based replication scheme which ensures the read-your-writes consistency model
- Results showed a reduction in Procedure Completion Times by up to 3.1x in cases of user mobility

Publications

Enabling Emerging Edge Applications Through a 5G Control Plane Intervention

Mukhtiar Ahmad, Ali Nawazish, M Taimoor Tariq, Basit Iqbal, Taqi Raza, Zafar Ayyub Qazi

ACM CoNEXT 2022, Rome, Italy

Neutrino: A Fast and Consistent Edge-based Cellular Control Plane

Mukhtiar Ahmad, Ali Nawazish, M Taimoor Tariq, Usman Jafri, Adnan Abbass, Mashal Abbas, Zartash Uzmi, Zafar Qazi

IEEE/ACM Transactions on Networking, 2022

Fast Failure Detectors for 5G Edge Deployments

M Taimoor Tariq, Maha Kamal, Vafa Batool

Student Research Competition, SOSP 2021

Teaching Experience

Teaching Assistant

August 2024 – Present

CS-425: Distributed Systems

UIUC, IL

Teaching Assistant

January 2022 – May 2022

CS-678: Topics in Internet Research (Graduate)

LUMS, Pakistan

Lead Teaching Assistant

September 2021 – December 2021

CS-582: Distributed Systems (Graduate)

LUMS, Pakistan

Teaching Assistant

January 2021 – May 2021

CS-382: Network-Centric Computing (Undergraduate)

LUMS, Pakistan